

Host A/B suite — netkit vs TF Lite

Apple Silicon · clean rebuild Jul 11, 2026 · warm-only · MLP batched 1000×10 · models + fixture images excluded from flash/RAM

Runtime flash / RAM (absolute)

MCU-style image only: netkit ELF TEXT/DATA minus fixture .o; TF Lite = core LiteRT CPU libs. Same TF footprint for both dtypes.

XNNPACK ON		~9.4× / 3.9× smaller		XNNPACK OFF		~64× / 48× smaller	
1.31–1.32 MiB	12.41 MiB	netkit flash	TF Lite flash	193–200 KiB	12.41 MiB	netkit flash	TF Lite flash
191.8 KiB	752.0 KiB	netkit RAM	TF Lite RAM	15.8 KiB	752.0 KiB	netkit RAM	TF Lite RAM

Warm-only methodology

Discard 1st process, keep 2nd; order swaps (nk→TF, TF→nk). MLP: discard batch pass 0. CNN: discard run 0 + image 0 (runs=9 both sides). ImageNet: warm_mean discards the full first image pass. Ratio = TF ÷ netkit (>1 ⇒ netkit faster / smaller).

INT8 — latency (absolute + TF÷netkit)

model	XNN	netkit	TF Lite	latency	flash	ram
mlp	ON	1.558 μs	1.140 μs	0.73×	9.4×	3.9×
cnn	ON	20.324 μs	21.270 μs	1.05×	9.4×	3.9×
cnn_dw	ON	20.765 μs	23.212 μs	1.12×	9.4×	3.9×
imagenet	ON	694 μs	721 μs	1.04×	9.4×	3.9×
mlp	OFF	2.883 μs	7.703 μs	2.67×	63.5×	47.7×
cnn	OFF	141 μs	551 μs	3.92×	63.5×	47.7×
cnn_dw	OFF	188 μs	450 μs	2.39×	63.5×	47.7×
imagenet	OFF	7.53 ms	28.5 ms	3.78×	65.6×	47.7×

FLOAT32 — latency (absolute + TF÷netkit)

model	XNN	netkit	TF Lite	latency	flash	ram
mlp	ON	1.200 μs	2.244 μs	1.87×	9.4×	3.9×
cnn	ON	30.915 μs	29.854 μs	0.97×	9.4×	3.9×
cnn_dw	ON	29.034 μs	30.620 μs	1.06×	9.4×	3.9×
imagenet	ON	1.054 ms	1.100 ms	1.04×	9.4×	3.9×
mlp	OFF	19.9 μs	45.4 μs	2.28×	65.8×	47.7×
cnn	OFF	517 μs	1.13 ms	2.19×	65.8×	47.7×
cnn_dw	OFF	256 μs	405 μs	1.58×	65.8×	47.7×

model	XNN	netkit	TF Lite	latency	flash	ram
imagenet	OFF	32.5 ms	60.9 ms	1.87×	63.6×	47.7×

Green rows: netkit ahead on latency (ratio ≥ 1). Amber: TF Lite ahead. Source: benchmark/host_ab_suite_results_int8.txt and host_ab_suite_results_float32.txt.